

[ 813 CHALLENGE // URBAN EXPANSION, LAND USE CHANGE & HEAT RISK ]

# CoolCairo

## Which blocks should Cairo cool first?

Summer heat risk mapped block by block from satellites, and a 3D tool where planners paint cool roofs, trees, cool pavements and parks and see the effect live.

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**TEAM 46**

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EGYPT



# Cairo's surfaces pass 45 °C every summer

45.7 °C

a typical east-Cairo block's summer surface temperature (Landsat, 2023–25)

95,141

residents in our 2.2 × 2.2 km Nasr City district (WorldPop 2024)

+1 °C

new blocks built 2016–23 run hotter than established ones (46.9 vs 46.0 °C)

**Who is affected:** residents of dense blocks on hot surfaces, old and new. Heat hits older people, young children and outdoor workers hardest.

We want to **map** summer heat risk (surface temperature, the materials driving it, and the residents exposed) in **fast-growing MENA cities**, starting with **Nasr City, Cairo**, so that **city planners** can decide which blocks to prioritise for cool roofs, street trees, cool pavements and pocket parks.

# End user: city planners cooling the blocks where people live

## 01 SEE

### Where is it hot?

Surface heat and surface materials per 90 m block, in 3D.

## 02 RANK

### Who is exposed?

Heat risk: residents × °C above a typical block.

## 03 PAINT

### Try a measure

Cool roofs, street trees, cool pavements, pocket parks, painted onto blocks.

## 04 COMPARE

### What changes?

Live before → after: temperature, heat exposure, residents cooled.

## WHO ADOPTS IT

City and governorate planning departments, new-city developers, and climate-adaptation programmes that fund cooling.

## WHY IT MATTERS

Cooler surfaces where people live, for health and the environment: a third of the blocks, chosen by risk, removes 74% of heat exposure.

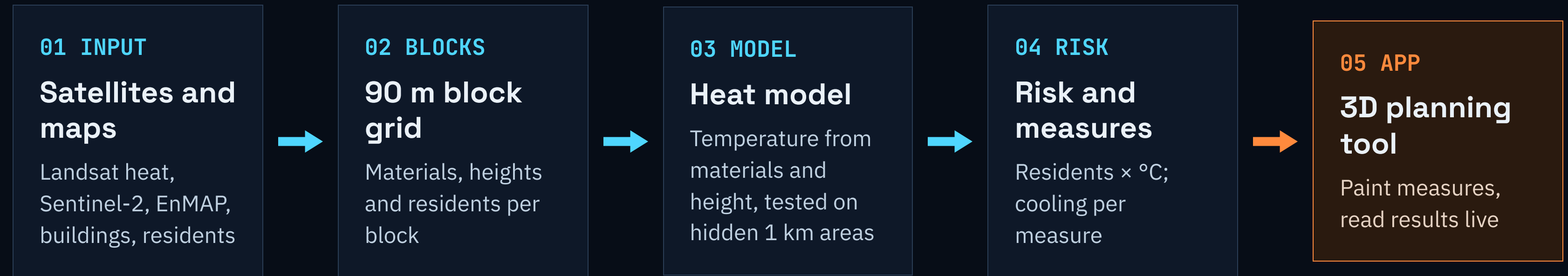
# Free satellite data that covers the whole region

| Source                                    | What we took                       | Used for                              |
|-------------------------------------------|------------------------------------|---------------------------------------|
| Landsat 8/9 (USGS)                        | 65 summer scenes, 2023–25          | Surface temperature, median per block |
| Sentinel-2 L2A (ESA)                      | 70 granules (58 days), 10 m        | Surface materials, vegetation         |
| <b>EnMAP L2A (DLR), hyperspectral</b>     | <b>1 scene, 22 Apr 2025, 30 m</b>  | <b>Which surfaces drive heat</b>      |
| OpenStreetMap, Google Open Buildings 2.5D | ~1,900 buildings, heights, 2016–23 | 3D city, roofs, urban growth          |
| WorldPop 2024                             | 100 m population grid              | Residents per block                   |

Streamed from Microsoft Planetary Computer and DLR EOC Geoservice. No paid data; raw EnMAP data stays out of the repository (licence).

**Why EnMAP, not Satellite 813:** 813 imagery of a team's own area comes in the incubation phase (organizers, 1 Oct 2026), so the PoC uses EnMAP; 813 data for Cairo is our first incubation step.

# From satellite scenes to a planning tool



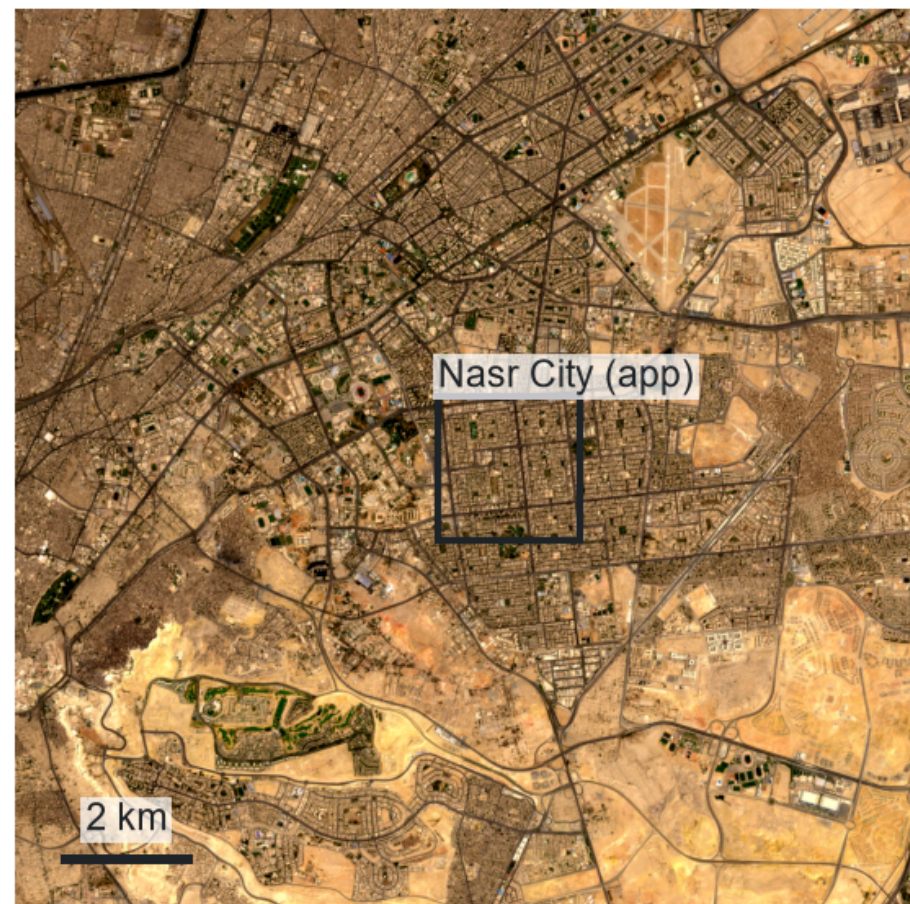
PYTHON 3.12 • uv • NOTEBOOKS 01-06 • UNIT TESTS

UNITY 6 APP

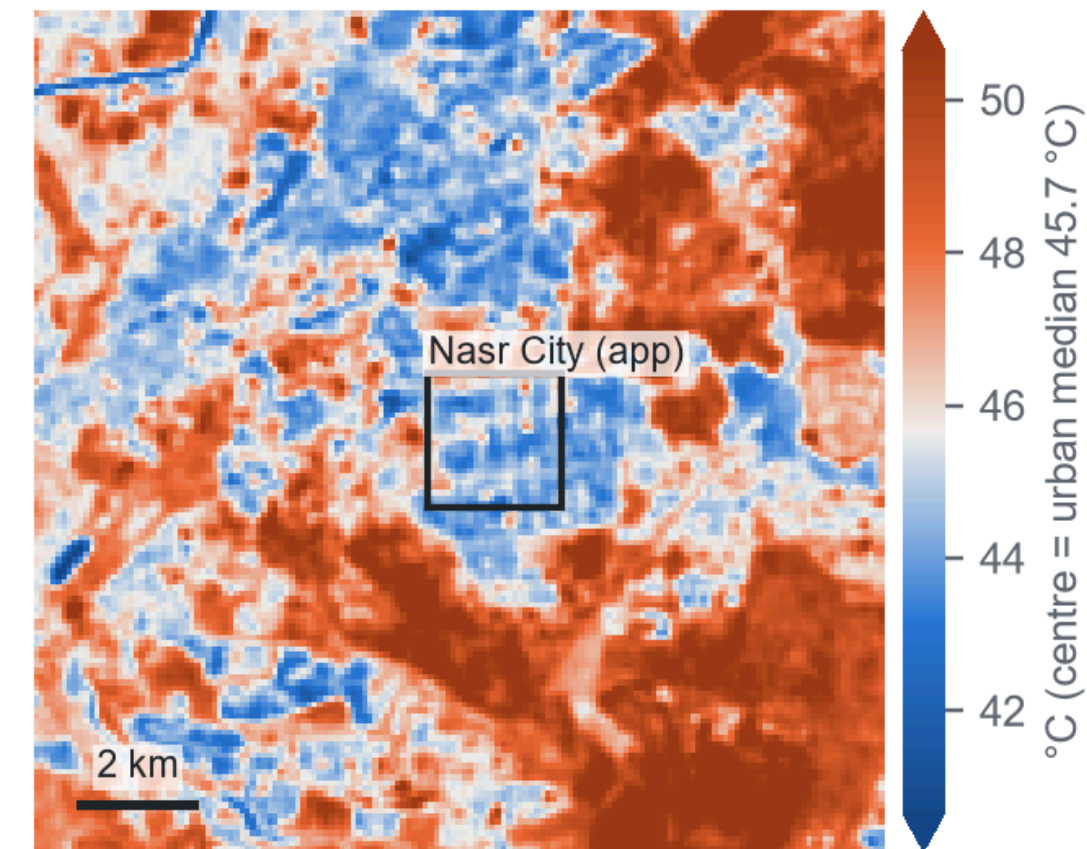
The analysis writes one file, [export/district.json](#); the app evaluates the same model live, so every number on screen traces back to a notebook.

# Desert hotter than the irrigated city

a Model area, Sentinel-2 true colour



b Summer surface temperature, 90 m blocks



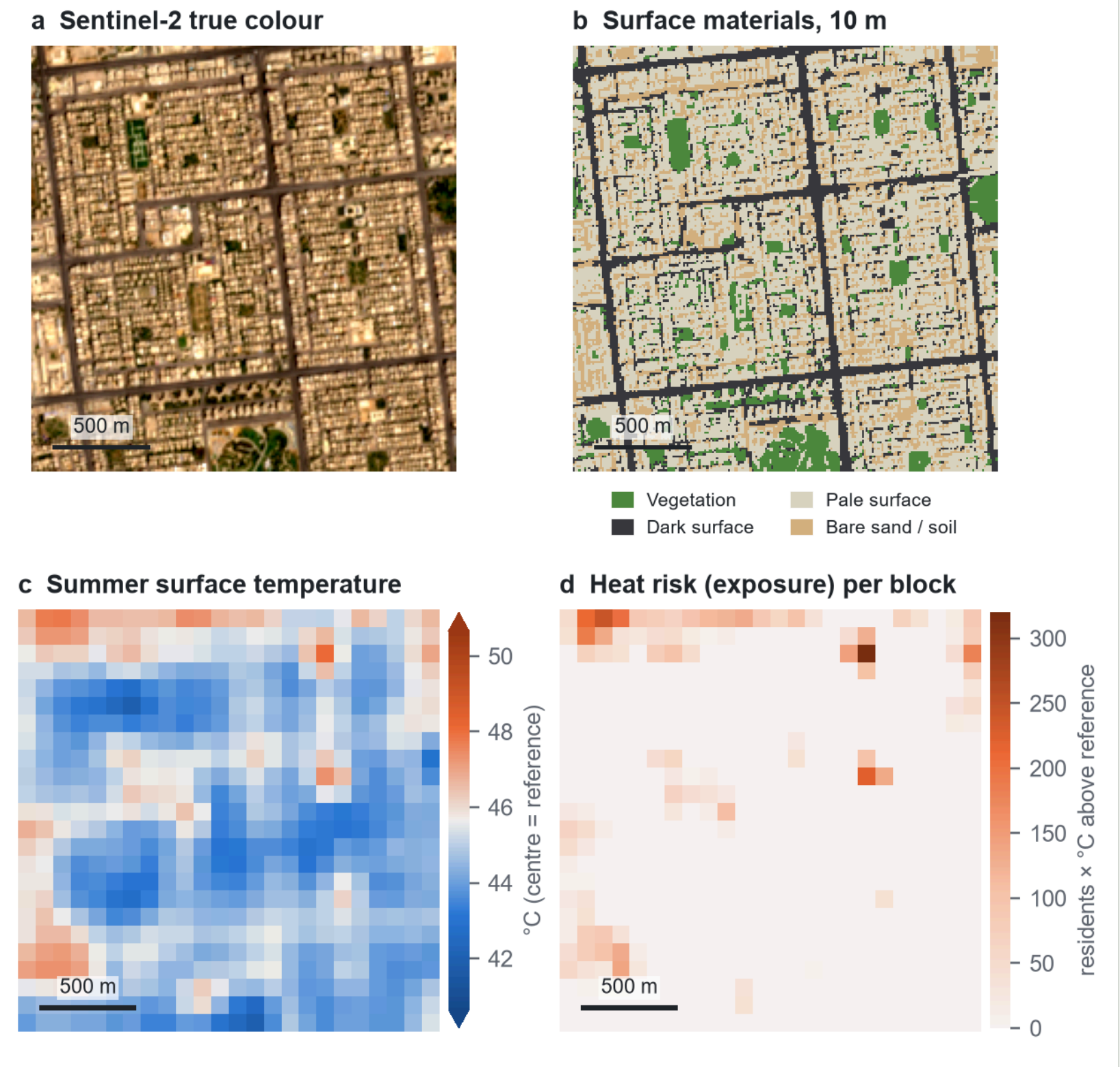
**(a)** Sentinel-2 summer photo; **(b)** Landsat surface temperature per 90 m block, median of 65 summer scenes, centred on the typical block (45.7 °C). Box: the Nasr City district in the app.



# Heat and heat risk are not the same map

Nasr City from the analysis: (a) Sentinel-2 summer photo, (b) surface materials at 10 m, (c) summer surface temperature per 90 m block, (d) heat risk, residents × °C above the typical block.

Bare desert edges are hottest, but few people live there.  
Risk concentrates where dense housing meets hot surfaces, so those blocks come first.



# Hyperspectral EnMAP explains heat best

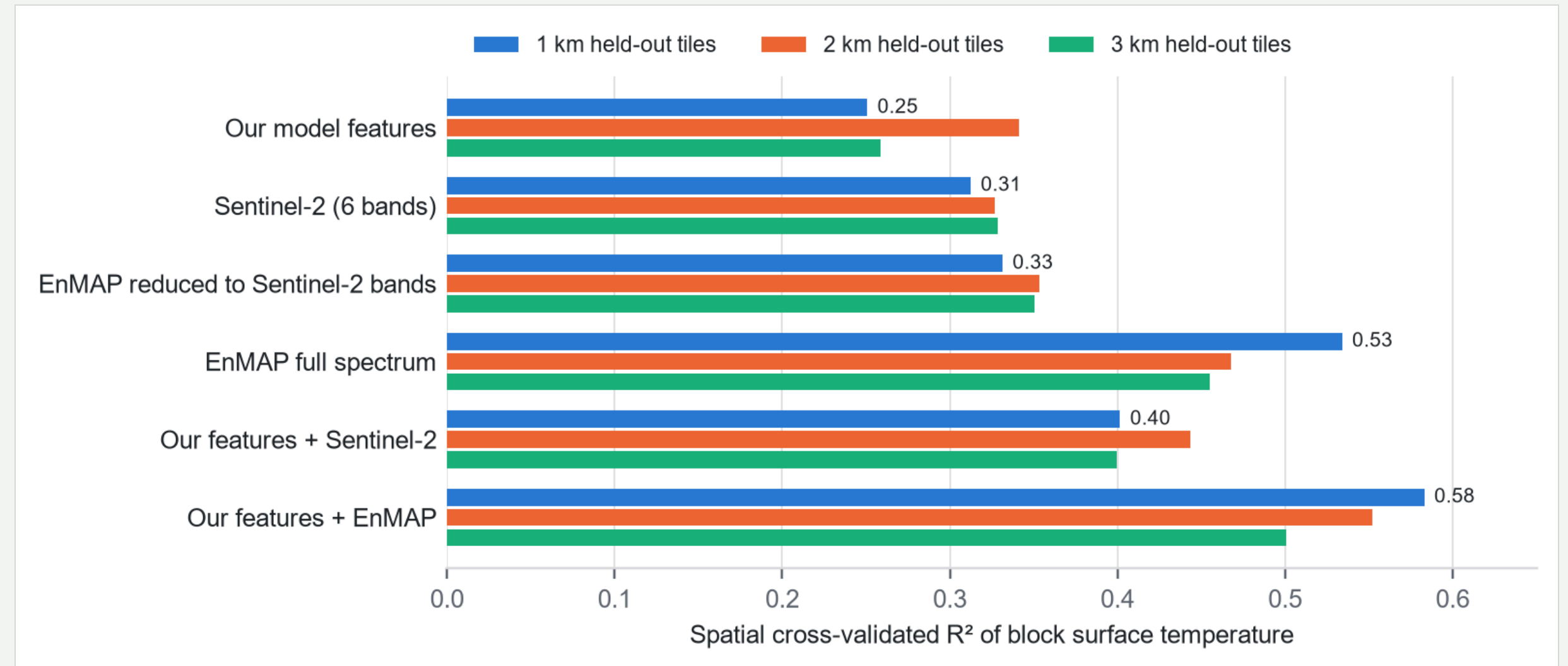
R<sup>2</sup> WITH OUR FEATURES

0.40

Sentinel-2 bands

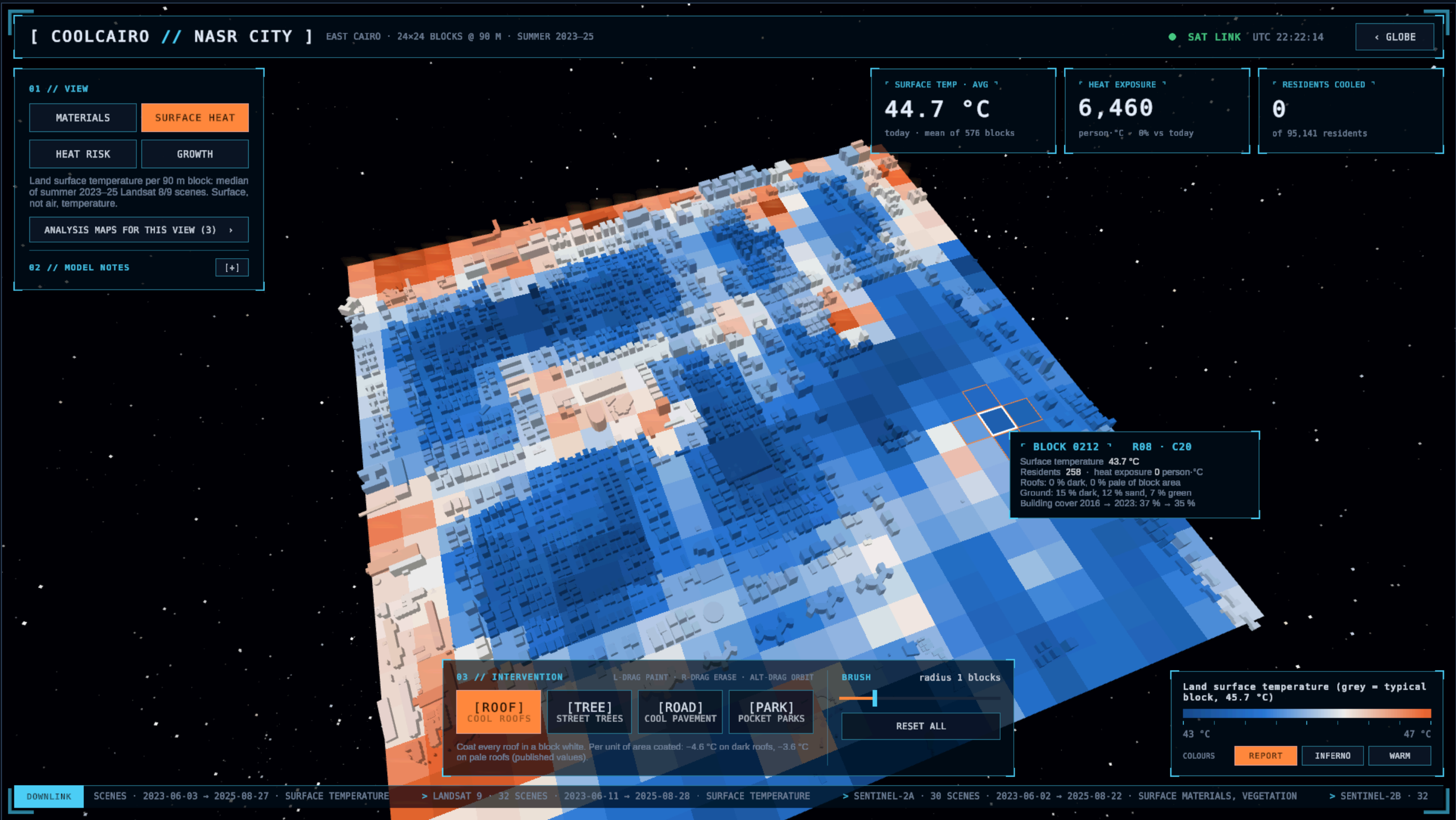
0.58

EnMAP hyperspectral



Hidden 1 km areas; holds at 2 and 3 km. Spectra alone: EnMAP 0.53 vs 0.33 for the same scene cut to Sentinel-2's bands, so the gain is spectral detail. No gain for telling buildings from desert.





09 // OUTPUT · THE  
TOOL, TODAY

44.7 °C

mean block surface  
temperature

6,460

heat exposure, person·°C

0

of 95,141 residents in cooled  
blocks

**The app today:** Surface heat view, summer surface temperature per 90 m block over 3D buildings. Left: views; bottom: the four cooling tools; right: the colour key.



10 // OUTPUT · THE TOOL, AFTER A PLAN

44.3 °C

mean block surface, was 44.7

1,686

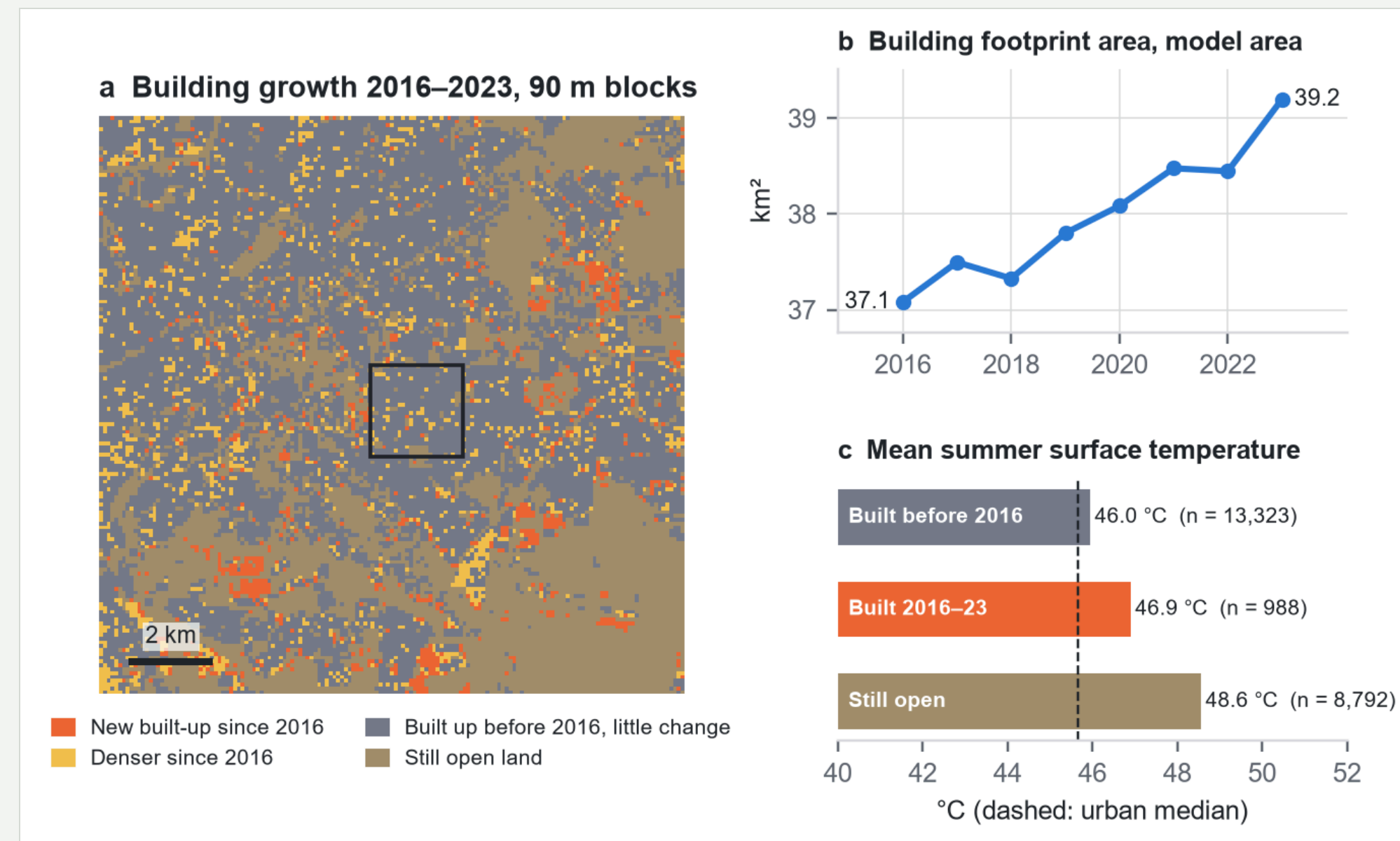
heat exposure, -74% vs today

22,588

residents in cooled blocks

**After a plan:** cool roofs and pocket parks painted on the riskiest third of blocks. Heat risk view: only a few blocks are still above the typical temperature.

# New neighbourhoods are being built hot



**+5.7%**

building footprint in east Cairo, 2016–23 (37.1 → 39.2 km²)

**988**

blocks turned from open land to built-up, now home to ~61,000 people

**46.9 °C**

new blocks, vs 46.0 established and 48.6 open desert



# What we checked, and what we cannot claim

**$R^2$  0.25**

Heat model on hidden 1 km areas,  $\pm 1.2$  °C, 2,538 blocks

**16 / 18**

material spot checks agree with satellite imagery (89%, not blind)

**$r$  0.60**

building heights vs OSM tags, bias  $-1.4$  m, error 6.2 m

**51 / 51**

automated checks pass inside the built app

**1 Surface, not air.** Satellites measure how hot surfaces get.

**3 Model explains ~a quarter.** Fine for comparing measures, not exact predictions.

**5 OpenStreetMap gaps.** The model learns from well-mapped blocks.

**7 One EnMAP scene (April).** A strong signal, not yet the main model.

**2 90 m blocks only.** The heat sensor cannot see single buildings.

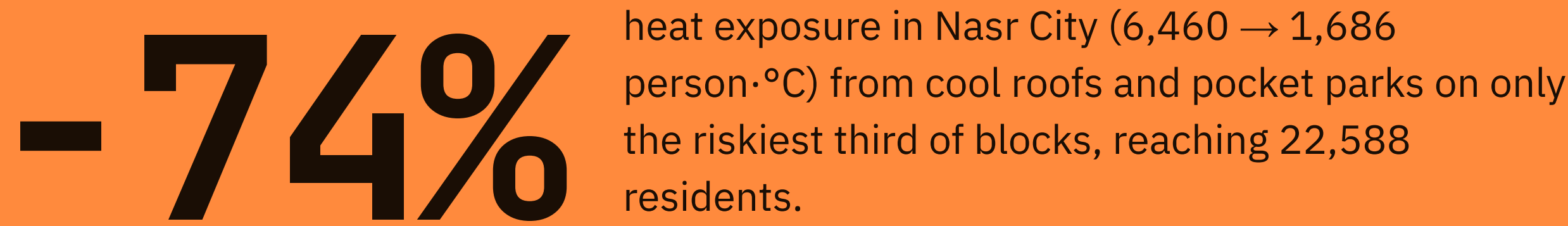
**4 Dusty roofs vs sand, shadows.** Why cool roofs use published physics.

**6 Modelled residents.** WorldPop; no age or housing vulnerability yet.

# How we would overcome each limitation

| Limitation                             | How to overcome it                                                                                   | Effort                        | Evidence it helps                       |
|----------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------|
| 1 Surface, not air temperature         | Weather stations on the same days; 10–20 air sensors for one summer; SOLWEIG comfort model           | 1–2 days; sensors: incubation | Direct test: hot surface = hot air?     |
| 2 Block level only (90 m)              | Sharpen to ~30 m with Sentinel-2; ECOSTRESS (~70 m, day and night); ESA LSTM, TRISHNA (~50 m)        | Days; missions when they fly  | Published; night heat matters most      |
| 3 Model explains ~25% ( $R^2$ 0.25)    | Add EnMAP spectra, neighbouring blocks, desert distance, street width; test a non-linear model       | Half a day                    | EnMAP measured: $R^2$ 0.40 → 0.58       |
| 4 Dusty roofs vs sand; shadows         | Classifier on all 6 Sentinel-2 bands; shadows from heights and sun angle; blind spot check           | About 1 day                   | Test pixels: 76% → 86%                  |
| 5 OpenStreetMap gaps in Cairo          | Google Open Buildings or Microsoft footprints as the roof map; add missing buildings to OSM          | 1–2 days                      | More than today's 2,538 training blocks |
| 6 Modelled residents; no vulnerability | WorldPop age grids (over-65s, under-5s), CAPMAS census, clinics and shade → heat vulnerability index | 1–2 days; index: incubation   | Standard public-health method           |
| 7 One EnMAP scene (April)              | Request summer EnMAP scenes; NASA EMIT, Tanager, PRISMA; Satellite 813 data in incubation            | Days to months                | One scene already: 0.40 → 0.58          |

# Targeting beats spreading: cool the right third of blocks



ONE MEASURE IN EVERY BLOCK, FOR COMPARISON



SDG FIT

**SDG 11** Sustainable cities

**SDG 13** Climate action

**SDG 3** Good health



# Heat-risk planning as a service

## CUSTOMERS

City and governorate planning departments, new-city developers, and climate-adaptation funders (development banks, NGOs) that pay for cooling programmes.

## PRODUCT AND DELIVERY

Today: the desktop app plus a reproducible pipeline. Next: a web version on GIQ. Each new city uses the same free data and code; deliverables are ranked block maps and costed cooling plans.

## REVENUE

A setup fee per city plus a yearly licence for updates and new summers; paid pilots; plan reports for funded cooling programmes.

## COSTS AND WHY IT PAYS

All input data is free (Landsat, Sentinel-2, OSM, WorldPop); costs are mainly team time and light cloud compute. For the client: in Nasr City, a third of the blocks removes 74% of heat exposure.

**First step:** a pilot with one Cairo district authority or new-city developer during incubation.

# From a Nasr City prototype to a MENA heat-planning tool

## Validate on the ground

10–20 air-temperature sensors in Nasr City for one summer; a blind check of the material map.

## See more

Summer hyperspectral scenes (EnMAP, EMIT, Satellite 813); night-time heat from ECOSTRESS.

## Map every building

Google Open Buildings and Microsoft footprints fill OpenStreetMap's gaps in Cairo.

## Protect the vulnerable

Add age (over-65s, under-5s) and access to cooling; a heat vulnerability index.

## Scale across MENA

Same free data and code for the next cities on our globe: Riyadh, Amman, Casablanca.

## Add costs

Price per block for each measure from Egyptian supplier quotes, so planners compare °C per pound.

01 // VIEW

MATERIALS

SURFACE HEAT

HEAT RISK

GROWTH

Land surface temperature is 30 m above ground level of summer 2023-25 Landsat data. It is not air temperature.

ANALYSIS MAPS FOR THIS VIEW (3) &gt;

02 // MODEL NOT

SURFACE TEMP · AVG

44.7 °C

today · mean of 576 blocks

HEAT EXPOSURE

6,460

person · °C · 0% vs today

RESIDENTS COOLED

0

of 95,141 residents

[ COOLCAIRO // TEAM 46 · EGYPT ]

# Thank you

## Liquaa Mahmoud

**TEAM LEAD**

GIS (ArcGIS Pro), 3D production, visual design, business case

## Mohamed Esmat

**TECHNICAL LEAD**

Satellite-data pipeline in Python, heat model, Unity 3D app

## Mahmoud Abu Zaid

**DATA & HYPERSPECTRAL ANALYST**

Earth-observation data, EnMAP vs Sentinel-2 hyperspectral analysis

Code, notebooks, methodology report and the Windows app: [github.com/mohamedEsmat/CoolCairo](https://github.com/mohamedEsmat/CoolCairo)

Data: Landsat (USGS) · Contains modified Copernicus Sentinel data [2023–2025] · Contains modified EnMAP data © DLR [2025] · WorldPop 2024 (CC BY 4.0) · © OpenStreetMap contributors · Google Open Buildings 2.5D Temporal (CC BY 4.0) · NASA Blue Marble

03 // INTERVENTION

L DRAG PAINT · R DRAG ERASE · ALT DRAG ORBIT

BRUSH radius 1 blocks

[ROOF]

COOL ROOFS

[TREE]

STREET TREES

[ROAD]

COOL PAVEMENT

[PARK]

POCKET PARKS

Coat every roof in a block white. Per unit of area coated: -4.6 °C on dark roofs, -3.6 °C on pale roofs (published values)

RESET ALL

Land surface temperature (grey = typical block, 45.7 °C)

43 °C

47 °C

COLOURS

REPORT

INFERNO

WARM